“I’m not! A much more respectable objection, to be honest, is that Many-Worlds is too lean and mean, and it’s therefore a nontrivial task to map the formalism onto the messiness of our observed world.”

Her father seemed to contemplate this. His cocktail sat temporarily neglected.

Alice decided to press the point. “I’ll explain what I mean. If you believe that quantum mechanics is saying something about reality, you believe that an electron can be in a superposition of spin-up and spin-down, for example. And since you and I and our measuring apparatuses are made of electrons and other quantum particles, the simplest thing to assume—the thing that Occam’s razor would suggest that you do—is that you and I and our measuring apparatuses can also be in superpositions, and indeed that the whole universe can be in superpositions. That is what is straightforwardly implied by the formalism of quantum mechanics, like it or not. It’s certainly possible to think about complicating the theory in various ways to get rid of all those superpositions or render them unphysical, but you should imagine William of Occam looking over your shoulder, tut-tutting with disapproval.”

“Seems like a bit of sophistry to me,” her father grumbled. “Philosophizing aside, a bunch of in-principle-unobservable parts of your theory doesn’t seem very simple at all.”

“Nobody can deny that Many-Worlds involves, you know, many worlds,” Alice conceded. “But that doesn’t count against the simplicity of the theory. We judge theories not by the number of entities they can and do describe but by the simplicity of their underlying ideas. The idea of the integers—‘-3, -2, -1, 0, 1, 2, 3 . . .’—is much simpler than the idea of, I don’t know, ‘-342, 7, 91, a billion and three, the prime numbers less than 18, and the square root of 3.’ There are more elements in the integers—an infinite number of them—but there is a simple pattern, making this infinitely big set easy to describe.”

“Okay,” said her father. “I can see that. There are a lot of worlds, but there is a simple principle that generates them, right? But still, by the time you actually have all those worlds, it must take an enormous amount of mathematical information to describe all them. Shouldn’t we be looking for a simpler theory where they just aren’t needed at all?”

“You’re welcome to look,” replied Alice, “and people certainly have. But by getting rid of the worlds, you end up making the theory more